Multiple Features & Overfitting

Yaya Wihardi, S.Kom., M.Kom.

Email: yayawihardi@upi.edu

Department of Computer Science Education

Universitas Pendidikan Indonesia

Outline

- Multiple Features
- Features Map
- Overfitting
- Regularization

Multiple Features

- Pada permasalahan di dunia nyata, fitur yang digunakan seringkali lebih dari satu, utk prediksi harga rumah: ukuran, jml kamar tidur, jml lantai, … → $x_1, x_2, x_3, \dots, x_n$

Size (feet²)	Number of bedrooms	Number of floors	Age of home (years)	Price ($1000)
2104	5	1	45	460
1416	3	2	40	232
1534	3	2	30	315
852	2	1	36	178
…	…	…	…	…

Features Mapping

- Pada permasalahan multiple features, model hipotesis merupakan pemetaan dari kombinasi fitur-fitur yang ada → feature map
- Contoh fungsi hipotesis untuk n buah fitur:

$$h_w(x) = w_0 + w_1 x_1 + w_2 x_2 + \cdots + w_n x_n$$

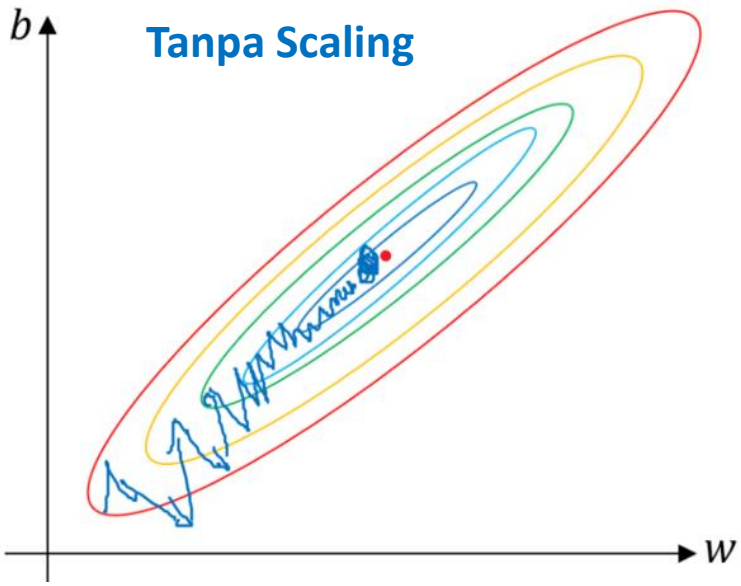
$$= w^T x$$ → **Utk x, 1 buah data**

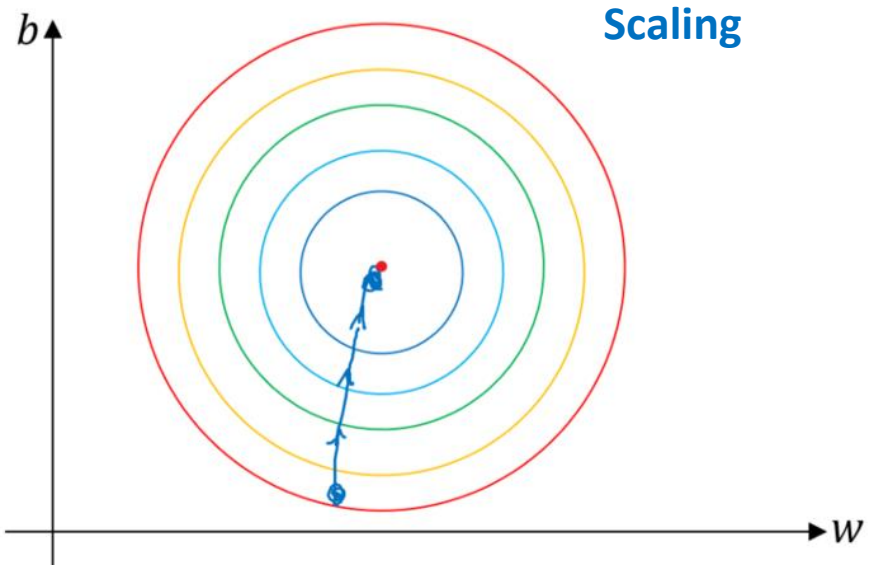
$$h_w(X) = X.w$$ → **Utk batch, X = m buah data**

$$x = \begin{bmatrix} 1 \\ x_1 \\ x_2 \\ \cdots \\ x_n \end{bmatrix} \quad w = \begin{bmatrix} w_0 \\ w_1 \\ w_2 \\ \cdots \\ w_n \end{bmatrix} \quad X = \begin{bmatrix} 1 & x_1^{(1)} & x_2^{(1)} & \cdots & x_n^{(1)} \\ 1 & x_1^{(2)} & x_2^{(2)} & \cdots & x_n^{(2)} \\ 1 & x_1^{(3)} & x_2^{(3)} & \cdots & x_n^{(3)} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_1^{(m)} & x_2^{(m)} & \cdots & x_n^{(m)} \end{bmatrix}$$

Bekerja dengan Multiple Features

- Masalah ketika bekerja dengan multiple features:
  - fitur memiliki skala berbeda
  - algoritma learning jadi tidak stabil → lambat
- Solusi: Scaling → Buat nilai setiap fitur dalam range yang sama, misal -1 s.d. 1 atau 0 s.d. 1
- Efek scaling:





Jenis Scaling

- Min-Max Scaling → tahu batas atas-bawah, data tidak normal

$$x' = \frac{x - x_{min}}{x_{max} - x_{min}}$$

range: 0 - 1

- Mean Normalization → data normal, robust terhadap outlier

$$x' = \frac{x - \mu}{x_{max} - x_{min}}$$

mean = 0

- Standardization → data normal, robust terhadap outlier

$$x' = \frac{x - \mu}{\sigma}$$

mean = 0 & stdev = 1

- Max Scaling

$$x' = \frac{x}{x_{max}}$$

Model Capacity

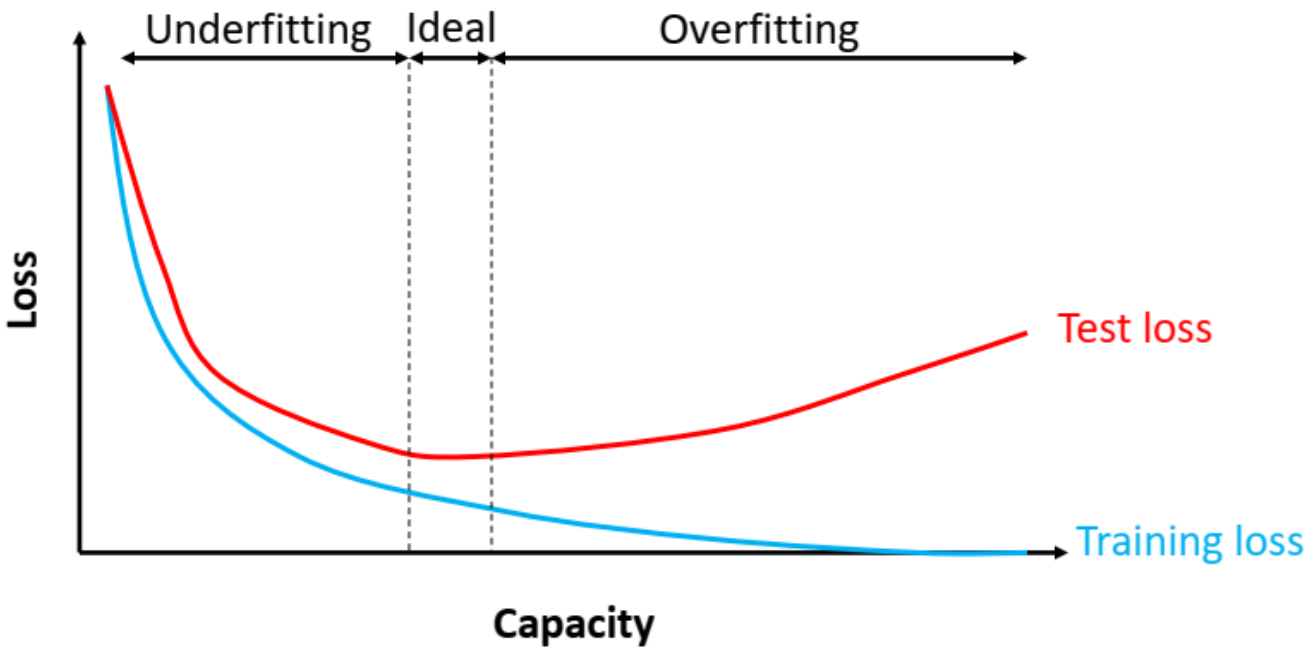
- Model capacity → kemampuan model mempelajari pola kompleks
- Model terlalu sederhana → underfitting
- Model terlalu kompleks → overfitting

Bias-Variance Tradeoff

- Bias = error karena model terlalu sederhana
- High bias → underfitting
- Variance = error karena model terlalu sensitif terhadap data
- High variance → overfitting
- Model yang baik harus menyeimbangkan bias dan variance.

Overfitting

- Kondisi Dimana model hipotesis mampu memodelkan data training dengan sangat baik, namun gagal melakukan generalisasi pada data baru yang belum pernah dilatihkan
- Training Loss → Kemampuan Memorization
- Test/Validation Loss → Kemampuan Generalization

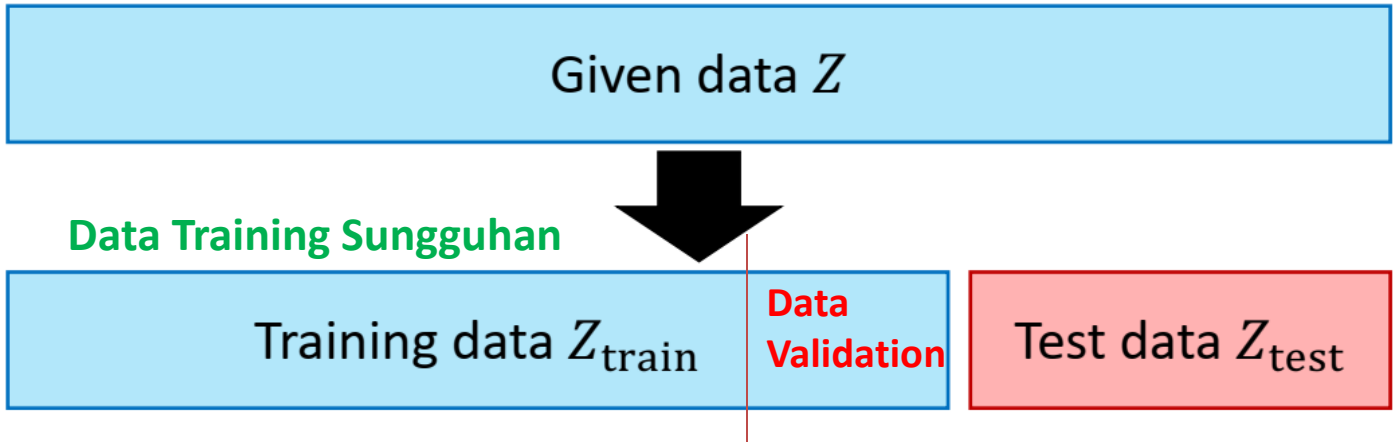


Masalah Dalam Multiple Features

- Overfitting → Jika fitur terlalu banyak kadang model hipotesis mampu memodelkan data training dengan sangat baik, namun gagal melakukan generalisasi pada data baru yang belum pernah dilatihkan
  - Solusi: Kurangi jml fitur, Regularization, Tambah Data
- Nonlinearity → Model hipotesis linear tidak mampu memodelkan data, sehingga dibutungkan model non-linear
  - Solusi: Gunakan model non-linear

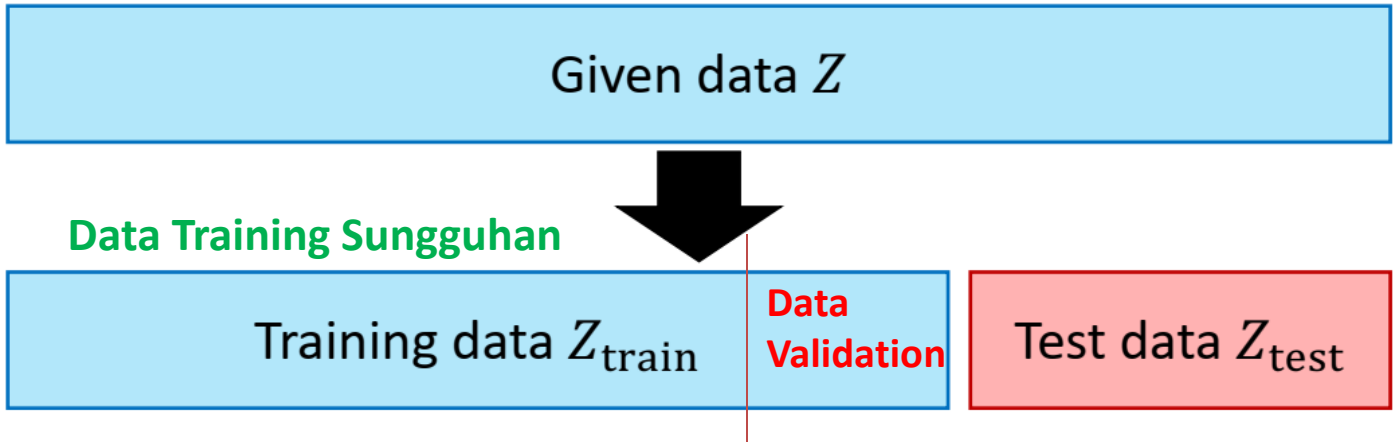
Deteksi Overfitting

- Bagaimana mendeteksi overfitting?
- Gunakan data splitting: Train-Test
- Sebelum displit, data dirandom terlebih dahulu, utk supervised learning, prosentase pembagian jml setiap kelas harus sama
- Model dilatih dengan data training, dan dievaluasi dengan menggunakan data testing.



Deteksi Overfitting

- Data Training → utk update bobot + Ngecek Memorization
- Data Validation → utk testing saat training → Ngecek Generalization saat training
- Data Validation tidak digunakan dalam perhitungan gradient
- Data Testing → utk evaluasi model akhir →Ngecek Generalization Model



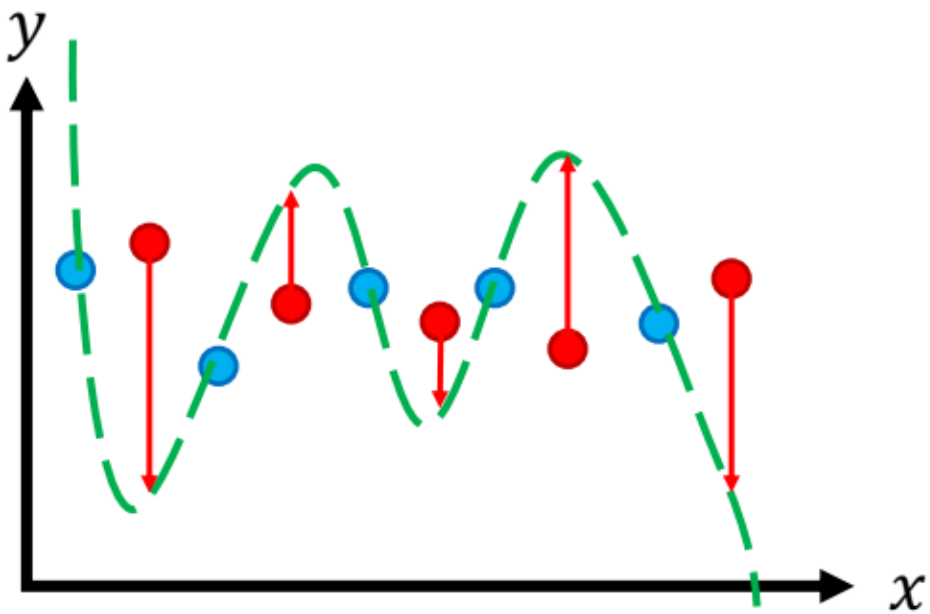
Deteksi Overfitting

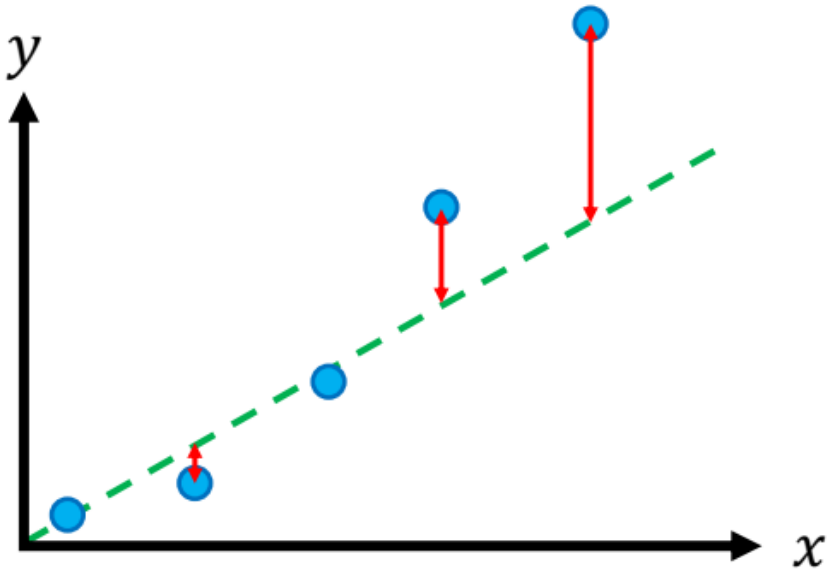
Overfitting

- Sangat Nge-Fit ke data training (Z_{train})
- Sangat jelek jika ditest dengan data Test (Z_{test})

Underfitting

- Kurang Nge-Fit ke data training (Z_{train})
- Sangat jelek jika ditest dengan data Test (Z_{test})





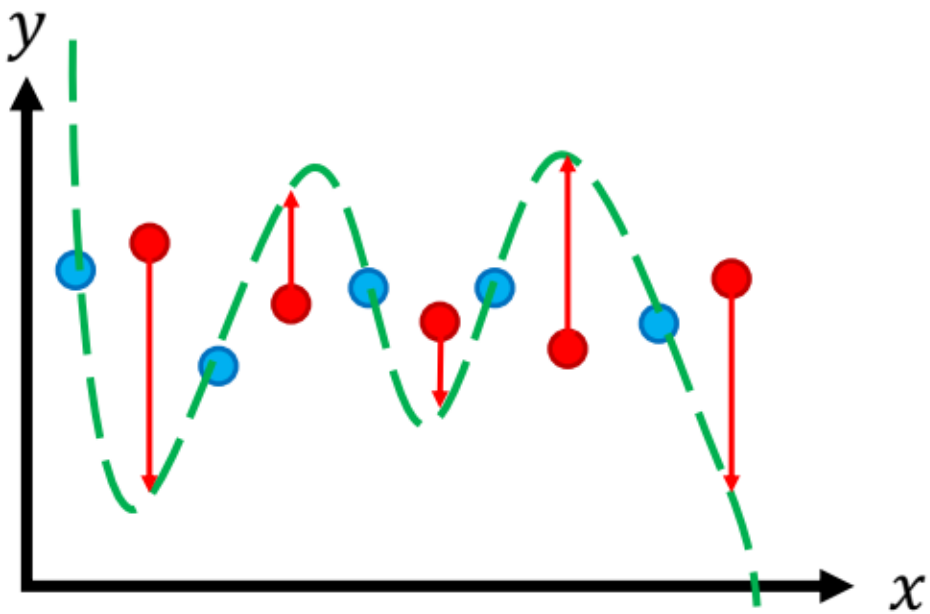
Deteksi Overfitting

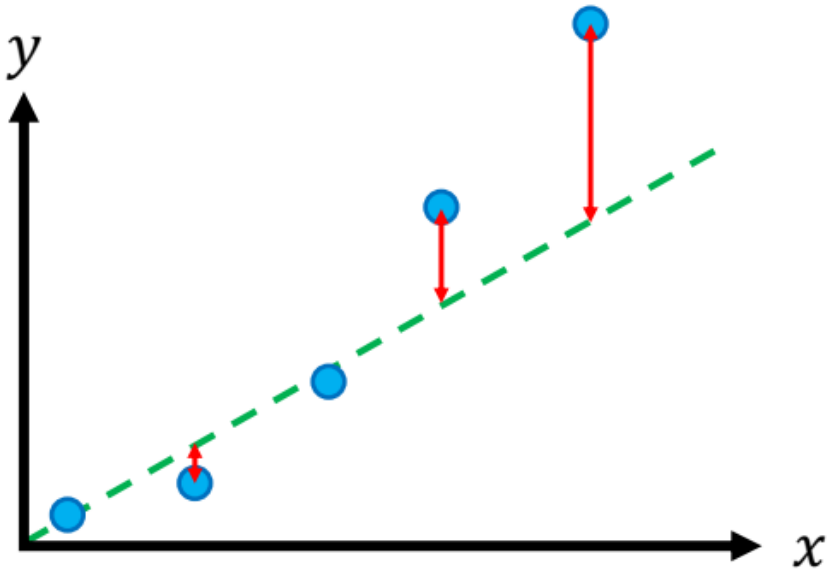
Overfitting

- Jml data latih (Z_{train}) sangat sedikit
- Jml data Test (Z_{test}) sangat banyak

Underfitting

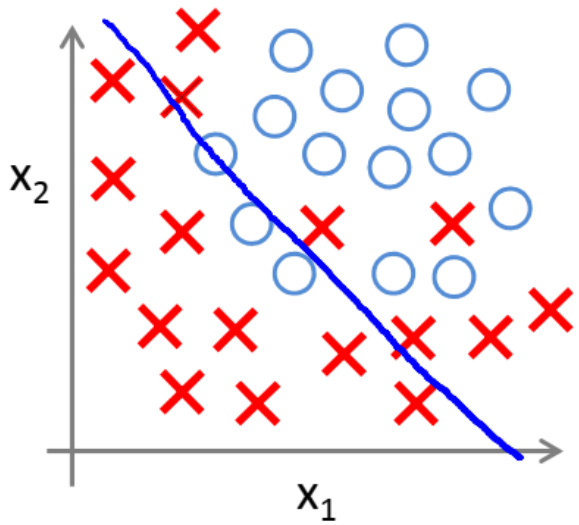
- Kurang Nge-Fit ke data training (Z_{train})
- Sangat jelek jika ditest dengan data Test (Z_{test})



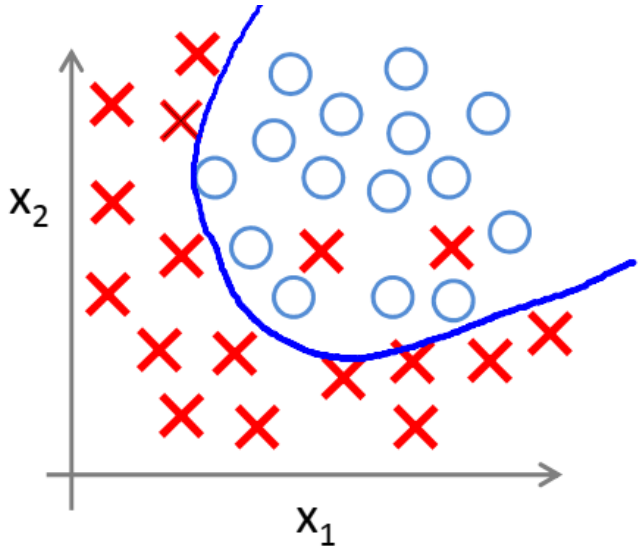


Deteksi Overfitting

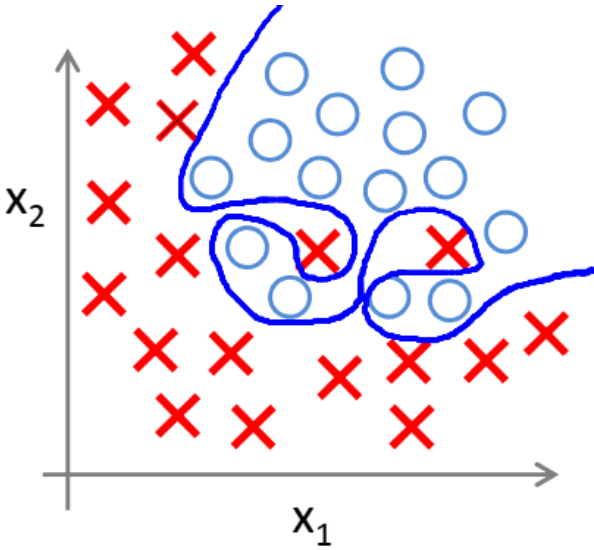
Underfitting



The Right Model



Overfitting



$$h_w(x) =$$

$$g(w_0 + w_1x_1 + w_2x_2)$$

$$g(w_0 + w_1x_1 + w_2x_2 + w_3x_1^2 + w_4x_2^2 + w_5x_1x_2)$$

$$g(w_0 + w_1x_1 + w_2x_2 + w_3x_1^2 + \cdots + w_6x_2^3 + w_7x_1^3x_2 + \cdots)$$

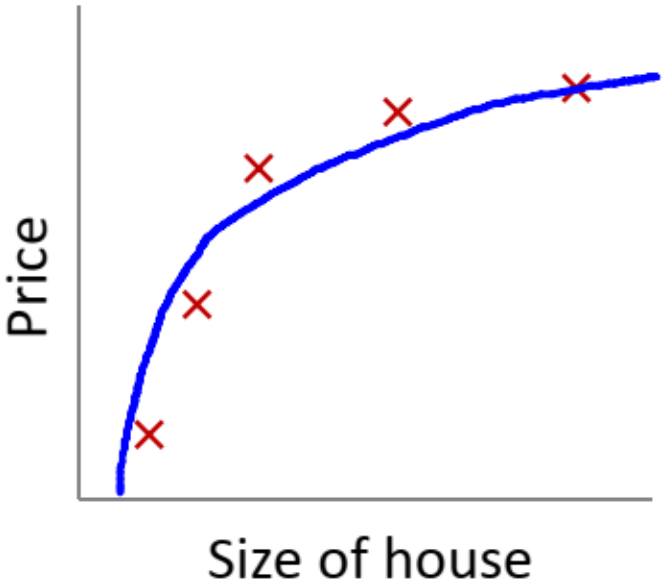
Contoh Overfitting

- Kasus: Credit Risk Model
  - Training accuracy: 98%
  - Test accuracy: 60%
- Penyebab:
  - model menghafal data training
  - fitur terlalu banyak
- Contoh lain?
  - Face recognition
  - Recommendation system
  - Fraud detection

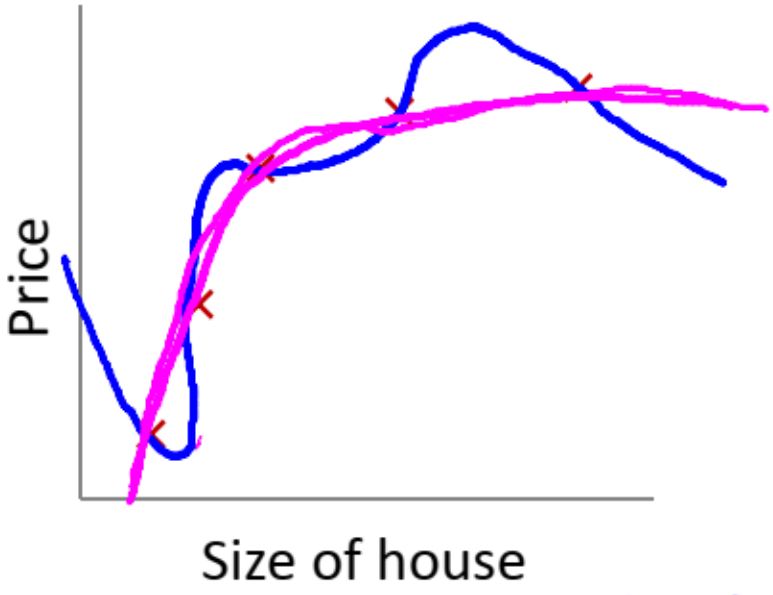
Solusi Overfitting

- Kurangi Jml Fitur: Features Selection → Sequential Feature Selection: Forward, Backward, dll
- Model Selection
- Regularization
- Tambah Jml Data

Regularization



$w_0 + w_1 x + w_2 x^2$



$w_0 + w_1 x + w_2 x^2 + w_3 x^3 + w_4 x^4$

- Idenya: Buat w_3 dan w_4 menjadi sangat kecil, caranya:

$$J(w_0, w_1) = \frac{1}{2m}\sum_{i=1}^{m}\left(h_w\left(x^{(i)}\right) - y^{(i)}\right)^2 + 1000\, w_3^2 + 1000\, w_4^2$$

Regularization

- Idenya: Buat w_3 dan w_4 menjadi sangat kecil, caranya:

$$J(w_0, w_1) = \frac{1}{2m}\sum_{i=1}^{m}(h_w(x^{(i)}) - y^{(i)})^2 + 1000\, w_3^2 + 1000\, w_4^2$$

- Nilai w yang besar akan memberi loss yang besar, sehingga model akan memilih w yang kecil
- Bentuk Umum

$$J(w_0, w_1) = \frac{1}{2m}\sum_{i=1}^{m}(h_w(x^{(i)}) - y^{(i)})^2 + \lambda \sum_{j=1}^{n} w_j^2$$

Thank You